Early and Long Term Colchicine After Acute Myocardial Infarction: CLEAR SYNERGY OASIS 9

Sanjit Jolly MD, MSc on behalf of CLEAR investigators

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HEALTH THROUGH KNOWLEDGE

Hamilton Health Sciences

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Disclosure of Relevant Financial Relationships

Within the prior 24 months, I have had a financial relationship with a company producing, marketing, selling, re-selling, or distributing healthcare products used by or on patients:

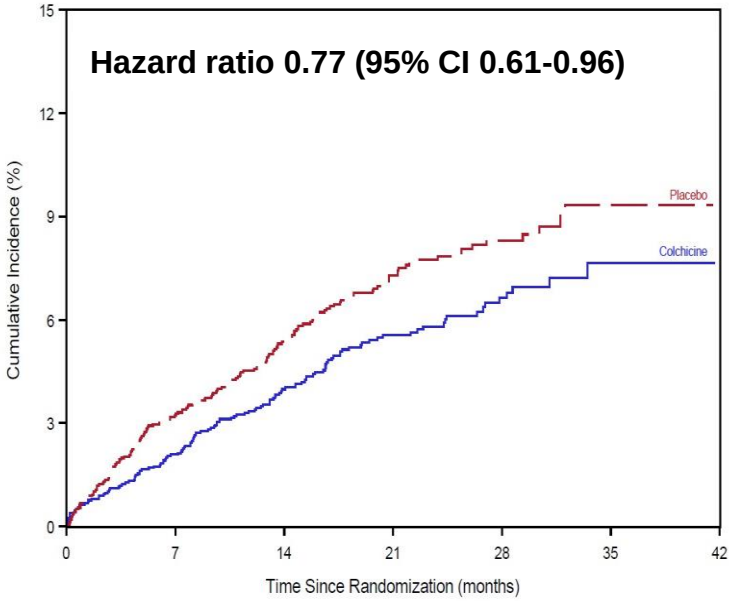
Nature of Financial Relationship	Ineligible Company
Grant/Research Support	Boston Scientific
Consultant Fees/Honoraria	Teleflex, Asahi, Shockwave, Abiomed
Individual Stock(s)/Stock Options	None
Royalties/Patent Beneficiary	None
Executive Role/Ownership Interest	None
Other Financial Benefit	None

CLEAR SYNERGY OASIS 9 Funding

- Funded by grants CIHR, PHRI and Boston Scientific

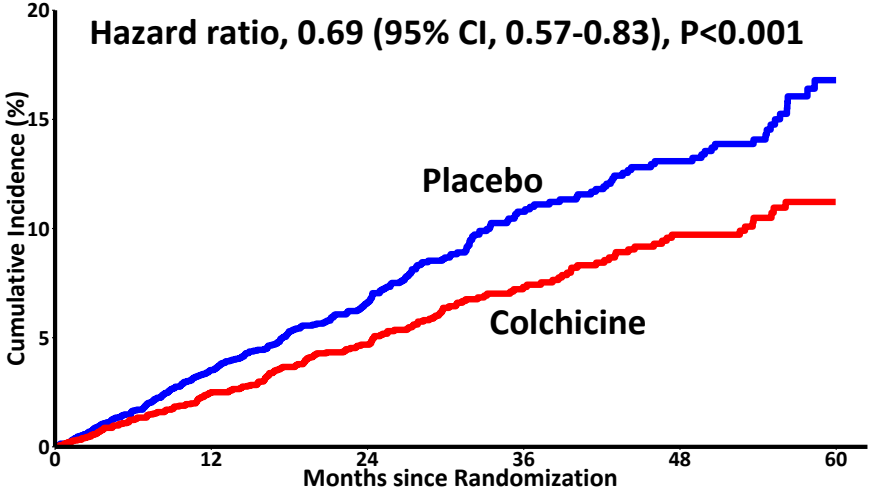
Two Recent Trials Showing Benefit for Colchicine in CAD

COLCOT- Acute MI (N=4745, 301 events)



No. at Risk							
Colchicine	2366	2284	1868	1230	628	153	0
Placebo	2379	2261	1854	1224	622	144	0

LODOCO2- stable CAD (N=5522, 451 events)

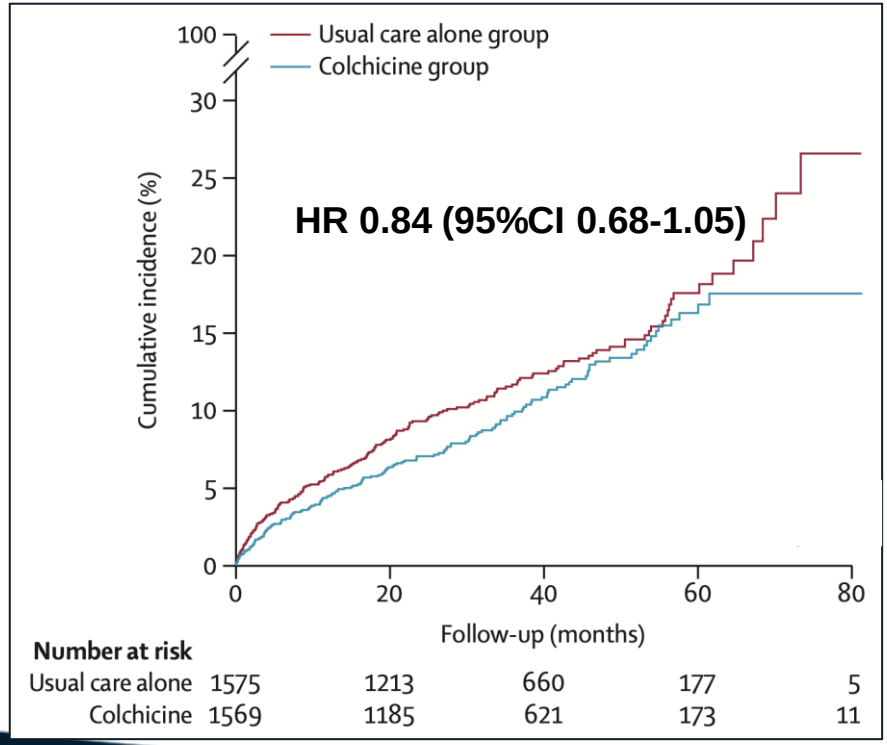


No. at Risk					
2760	2655	1703	821	590	161
2762	2685	1761	890	629	166

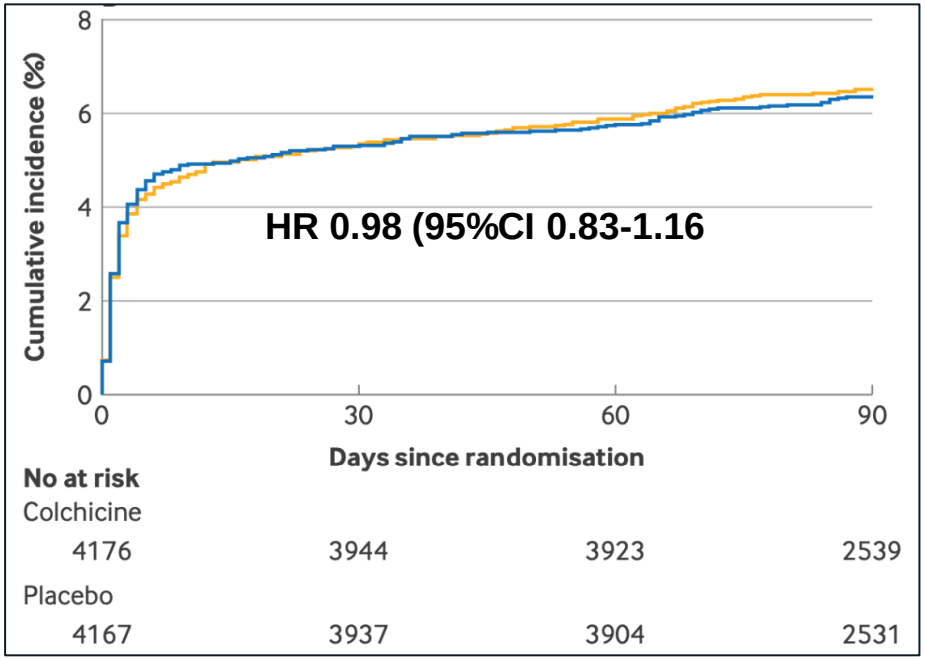
Tardif JC, et al. NEJM. 2019;381:2497-2505.
Nidorf M, et al. NEJM 2020; 5;383(19):1838-1847.

Cerebrovascular Disease: No Difference

CONVINCE (N = 3154, MACE+, 338 events)



CHANCE3 (N = 8343, all stroke, 534 events)



Kelly, P, et al. Lancet 2024; 404: 125-133.
Li, J, et al. BMJ 2024; 385.

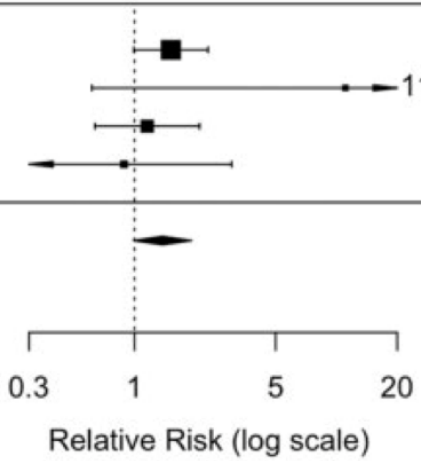
Does Colchicine Increase Non-CV Death?

Meta-analysis

Non-cardiovascular death

Acronym or author	Colchicine Events/total (%)	Comparator Events/total (%)	Weight (GIV)	Relative Risk (95% CI)
LoDoCo2, 2020	53/2762 (1.9%)	35/2760 (1.3%)	60.6 %	1.51 (0.99 to 2.31)
COPS, 2020	5/396 (1.3%)	0/399 (0.0%)	1.3 %	11.08 (0.61 to 199.77)
COLCOT, 2019	23/2366 (1.0%)	20/2379 (0.8%)	30.8 %	1.16 (0.64 to 2.10)
LoDoCo, 2013	5/282 (1.8%)	5/250 (2.0%)	7.3 %	0.89 (0.26 to 3.03)
Totals:	86/5806	60/5788	100 %	1.38 (0.99 to 1.92)

$I^2 = 0.4\%$. Random effects model for overall effect, $p = 0.060$



Rationale

- CLEAR trial started prior to results of COLCOT trial
- Larger trial with significantly more events and power
- Replication of prior results important for Class I indications

Primary Objective

- In patients with STEMI or large NSTEMI:
- Does a routine colchicine compared to placebo long term reduce the composite of CV death, MI, stroke or ischemia driven revascularization

CLEAR OASIS 9 Trial

7000 patients diagnosed with Acute Myocardial Infarction (MI) referred for PCI

↓

SYNERGY stent recommended for use when available*

↓

Randomized within 72 hours of PCI 2x2 Factorial

- Colchicine
  - Spironolactone
  - Placebo
- Placebo
  - Spironolactone
  - Placebo

Randomized

Primary Outcome

Colchicine vs. placebo: Composite of CV death, MI, stroke or ischemia driven revascularization over duration of follow-up

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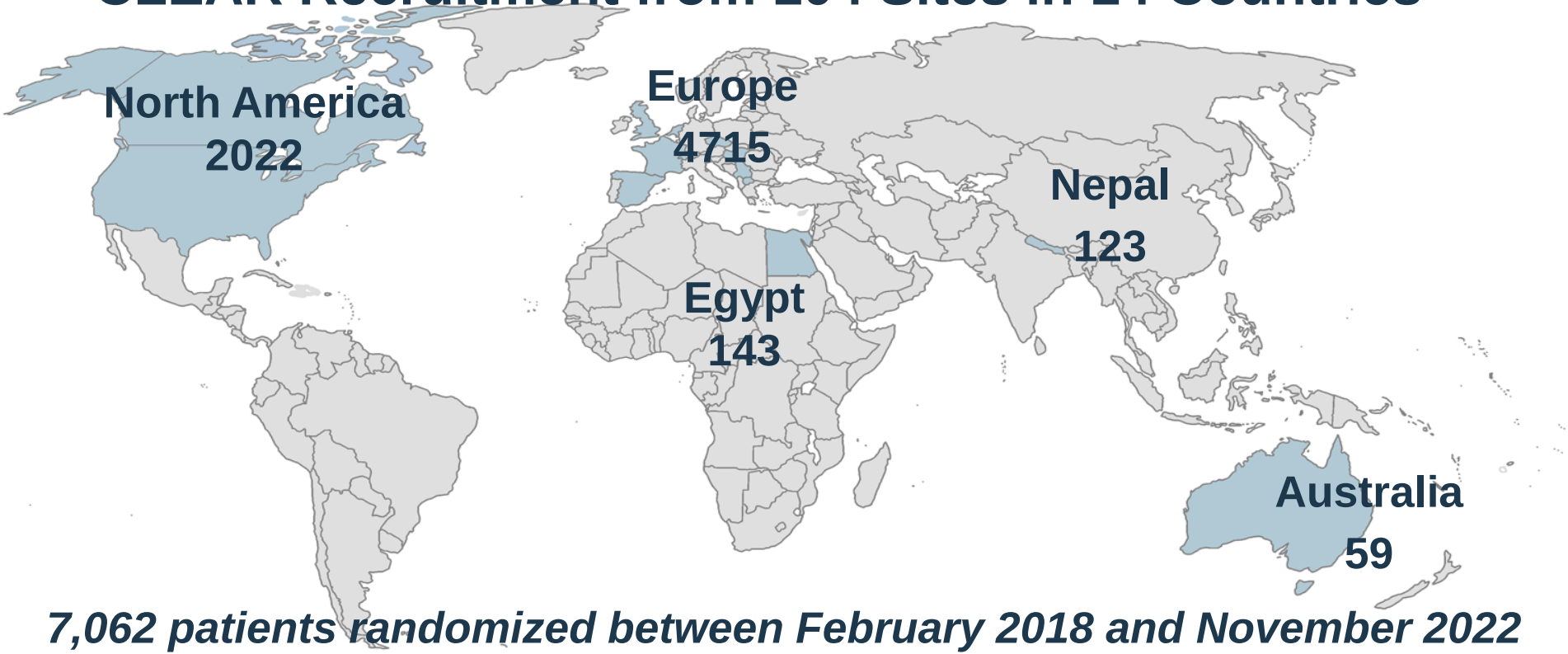
CLEAR Recruitment from 104 Sites in 14 Countries



7,062 patients randomized between February 2018 and November 2022

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Study Power and Follow Up

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- **Study power:** 80% power for a 25% RRR for a 9% control event rate
- **Analysis:** Intention-to-treat Cox proportional hazards model, stratified by STEMI vs. NSTEMI and spironolactone vs. placebo, accounting for competing risk
- **Sensitivity:** On treatment analysis
- **Follow up:** 99.4% in both groups

Baseline Characteristics

	Colchicine N=3528	Placebo N=3534
Mean Age	60.6	60.7
Female	20.5%	20.2%
STEMI	95.3%	94.8%
NSTEMI	4.7%	5.2%
Diabetes	18.7%	18.3%
Prior MI	8.8%	9.2%

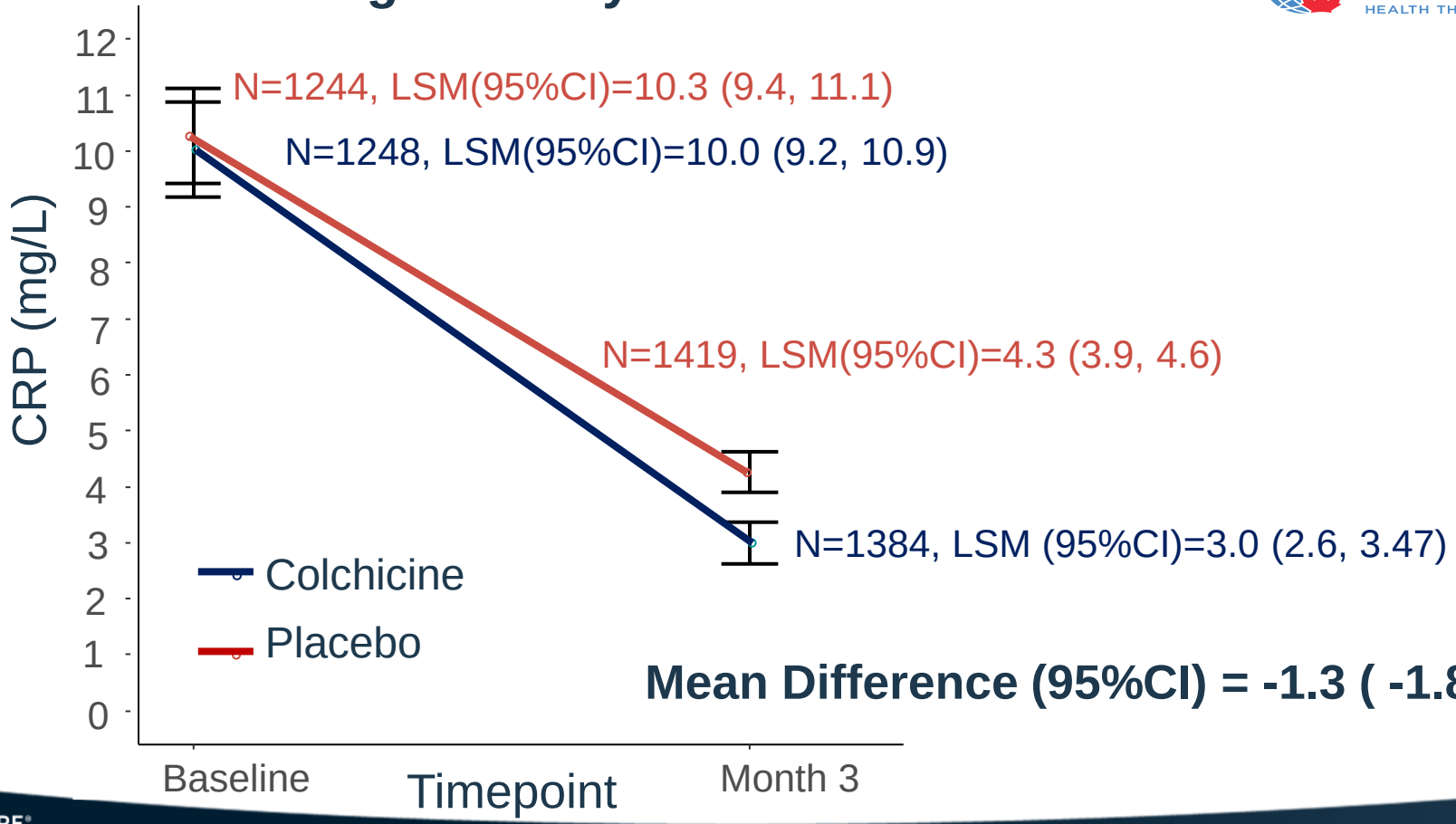
Medications at Discharge

	Colchicine N=3528	Placebo N=3534
Aspirin	97.2%	96.3%
Clopidogrel	41.9%	42.4%
Ticagrelor	45.7%	44.5%
Prasugrel	10.8%	11.7%
ACE or ARB	77.9%	78.3%
Statin	96.6%	96.7%
SLGT2 inhibitor	3.1%	2.9%

Initial PCI Procedure

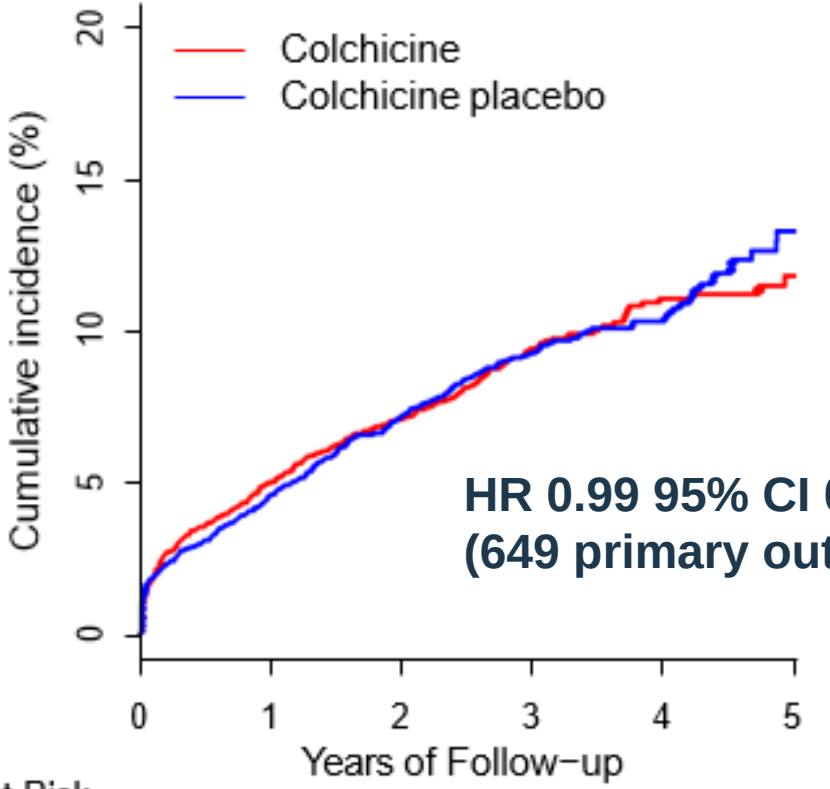
	Colchicine N=3528	Placebo N=3534
Bare metal stent	0.3%	0.2%
Drug-eluting stent	96.3%	95.8%
Angioplasty only	3.0%	3.4%
Number of stents (median)	1.0	1.0
Stent length (mm) mean	23.8	23.8
Stent diameter (mm) mean	3.2	3.1
Multivessel coronary disease	49.2%	49.3%

CRP was Significantly Reduced with Colchicine



Mean Difference (95%CI) = -1.3 (-1.8, -0.7)

Primary Outcome of CV Death, MI, Stroke of Ischemia Driven Revascularization



HR 0.99 95% CI 0.85-1.16, p=0.93
(649 primary outcome events)

No. at Risk						
Colchicine	3528	3329	2688	1686	697	183
Placebo	3534	3349	2683	1674	659	163

Primary Outcome

	Colchicine (N=3528) (%)	Placebo (N=3534) (%)	HR	95% CI	p
CV death, MI, stroke or ischemia driven revascularization	9.1%	9.3%	0.99	0.85-1.16	0.93
CV death	3.3%	3.2%	1.03	0.80-1.34	
MI	2.9%	3.1%	0.88	0.66-1.17	
Stroke	1.4%	1.2%	1.15	0.72-1.84	
Ischemia driven revascularization	4.6%	4.7%	1.01	0.81-1.17	
CV death, MI or stroke	6.8%	7.1%	0.98	0.82-1.17	
All cause death	4.6%	5.1%	0.90	0.73-1.12	
Non-CV death	1.3%	1.9%	0.68	0.46-0.99	

On Treatment Analysis

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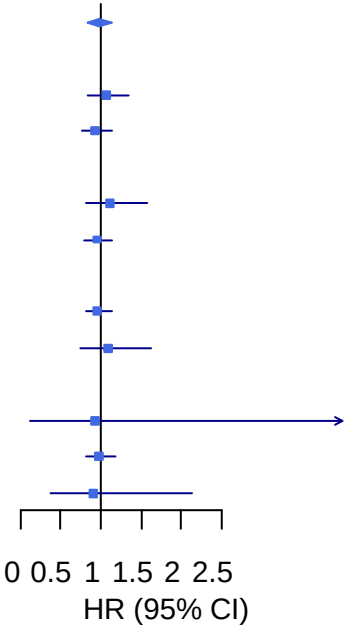
	Colchicine (N=3488) (%)	Placebo (N=3492) (%)	HR	95% CI	p
CV death, MI, stroke or ischemia driven revascularization	7.5%	7.5%	0.99	0.85-1.16	0.93
CV death	2.7%	2.5%	1.04	0.80-1.35	
MI	2.3%	2.4%	0.89	0.67-1.18	
Stroke	1.1%	1.0%	1.09	0.68-1.75	
Ischemia driven revascularization	3.9%	3.8%	1.03	0.82-1.29	
CV death, MI or stroke	5.5%	5.7%	0.97	0.81-1.16	
All cause death	3.4%	3.8%	0.90	0.70-1.15	
Non-CV death	0.7%	1.3%	0.71	0.49-1.04	

Adverse Events

	Colchicine (N=3528) (%)	Placebo (N=3534) (%)	p
Serious Adverse Events	6.7%	7.4%	0.22
Adverse Events	31.9%	31.7%	0.86
Serious Infection	2.5%	2.9%	0.85
Diarrhea	10.2%	6.6%	<0.001

Forest Plot of Primary Outcome in Pre-Specified Subgroups (I)

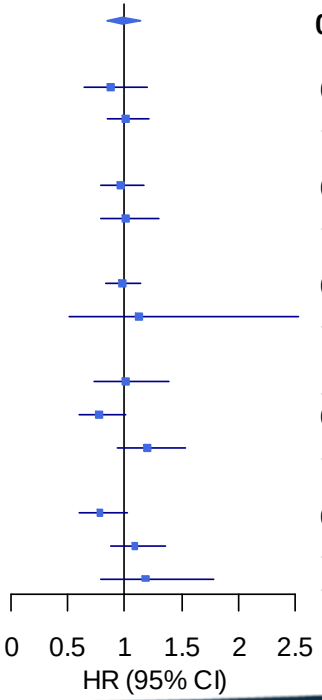
	Colchicine Count/Total (%)	Placebo Count/Total (%)	Hazard Ratio (95% CI)	P value for Interaction
Outcomes				
CVD/MI/Stroke/Revas	**322/3528(9.1)**	**327/3534(9.3)**	**0.98 (0.84-1.15)**	
Age				**0.418**
Age>= 65	140/1185(11.8)	135/1214(11.1)	1.06 (0.84-1.34)	
Age<65	182/2343(7.8)	192/2320(8.3)	0.93 (0.76-1.14)	
Sex				**0.384**
Female	72/725(9.9)	64/713(9.0)	1.12 (0.80-1.57)	
Male	250/2803(8.9)	263/2821(9.3)	0.95 (0.80-1.13)	
GFR level				**0.5**
GFR>=60	266/3237(8.2)	279/3256(8.6)	0.96 (0.81-1.13)	
GFR<60	56/291(19.2)	48/278(17.3)	1.10 (0.75-1.62)	
Region				**0.951**
North America	96/1010(9.5)	95/1012(9.4)	0.93 (0.11-7.66) *	
Europe	216/2356(9.2)	221/2359(9.4)	0.97 (0.81-1.17)	
Other region	10/162(6.2)	11/163(6.7)	0.90 (0.38-2.13)	



**HR at 1 year using a time-dependent Cox Model to account for the PH assumption violation.*

Forest Plot of Primary Outcome in Pre-Specified Subgroups (II)

	Colchicine Count/Total (%)	Placebo Count/Total (%)	Hazard Ratio (95% CI)	P value for Interaction
Outcomes				
CVD/MI/Stroke/Revas	**322/3528(9.1)**	**327/3534(9.3)**	**0.98 (0.84-1.15)**	
Diabetes				**0.432**
Yes	79/658(12.0)	85/645(13.2)	0.88 (0.65-1.20)	
No	243/2870(8.5)	242/2889(8.4)	1.01 (0.85-1.21)	
Single/Multi-vessel status				**0.745**
Multi	192/1735(11.1)	200/1742(11.5)	0.97 (0.79-1.18)	
Single	130/1793(7.3)	127/1792(7.1)	1.02 (0.80-1.30)	
MI Type				**0.741**
STEMI	310/3363(9.2)	315/3350(9.4)	0.98 (0.84-1.15)	
NSTEMI	12/165(7.3)	12/184(6.5)	1.13 (0.51-2.52)	
Dosing				**0.054**
Once daily (<70Kg)	74/721(10.3)	72/697(10.3)	1.01 (0.73-1.40)	
Twice daily (>=70kg)	110/1161(9.5)	136/1137(12.0)	0.78 (0.61-1.00)	
Once daily (>=70kg)	138/1646(8.4)	119/1700(7.0)	1.20 (0.94-1.54)	
Covid Phase				**0.098**
Pre-covid	100/998(10.0)	125/991(12.6)	0.78 (0.60-1.02)	
During covid	170/1773(9.6)	159/1799(8.8)	1.09 (0.88-1.35)	
Post- covid	52/757(6.9)	43/744(5.8)	1.19 (0.79-1.78)	



Conclusions

- Acute and long term colchicine did not reduce composite of CV death, MI, stroke or ischemia driven revascularization
- Colchicine was associated with an increase in diarrhea
- CLEAR is the largest trial of colchicine in acute MI with substantially more events than prior trials

Implications

- The role of colchicine post myocardial infarction long term is uncertain

Will be available online at www.NEJM.org soon

The NEW ENGLAND JOURNAL of MEDICINE

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